Py

Day 08/55 - String Manipulation

Concatenation, indexing, slicing, and text formatting

Prepared by Satyanarayan Sen

Video Summary

This video provides a practical introduction to string manipulation in Python. It covers joining strings, reading characters with positive and negative indexes, slicing substrings, why strings cannot be changed in place, and simple length and multiline formatting tools.

Source video: https://www.youtube.com/watch?v=z-RbovDmtW8&list=PLsyeobzWxl7omDoEYrrf3oXvXxa6MPgek&index=7

1 String Concatenation

Concatenation means joining strings together. Python can combine adjacent string literals automatically, and the + operator joins string values stored in variables.

+ Literal join

Place string literals side by side when the text is fixed.

+ + operator

Use + when values are stored in variables or built step by step.

_ Add spaces

Python joins exactly what you give it, so include spaces when needed.

```
Concatenation examples
>>> "Hello" "Python"
'HelloPython'
>>> first = "Hello"
>>> second = "Python"
>>> first + " " + second
'Hello Python'
```

2 Indexing

A string is a sequence of characters. Indexes start at 0 from the left, and negative indexes count backward from the end.

```
Indexing examples
>>> word = "Telusko"
>>> word[0]
'T'
>>> word[3]
'u'
>>> word[-1]
'o'
```

Indexing rule

Use positive indexes when counting from the beginning and negative indexes when the useful character is near the end.

0 Zero based

The first character is index 0, not index 1.

+ Forward

Positive indexes move from left to right through the text.

-1 Backward

Negative indexes move backward from the last character.

3 Slicing

Slicing extracts a smaller string from a larger string. In [start:end], start is included and end is excluded.

```
Slicing examples
>>> word = "Telusko"
>>> word[0:3]
'Tel'
>>> word[2:]
'lusko'
>>> word[-3:]
'sko'
```

Slice	Meaning	Result for word = "Telusko"
word[0:3]	Start at index 0, stop before index 3.	'Tel'
word[2:]	Start at index 2 and continue to the end.	'lusko'
word[:4]	Start at the beginning and stop before index 4.	'Telu'
word[-3:]	Take the last three characters.	'sko'

Slicing rule

The start index is included. The end index is excluded, so word[0:3] returns indexes 0, 1, and 2.

4 Immutability

Strings are immutable. After a string is created, Python will not let you change one character or replace a slice in place.

```
Immutability examples
>>> word = "Telusko"
>>> word[0] = "M"
TypeError: 'str' object does not support item assignment
>>> "M" + word[1:]
'Melusko'
```

Beginner rule

To change text, build a new string from the pieces you want. The original string remains unchanged.

5 Essential Functions

The len() function counts how many characters a string contains, including spaces and punctuation.

```
len() examples
>>> len("Python")
6
>>> len("Hello Python")
12
```

6 Multiline Strings and Formatting

Multiline text can be created with the newline escape sequence or with triple double quotes when the text should be easier to read.

\n **\n escape**

Use \n inside a string to move the next text to a new line.

"""" **Triple quotes**

Use """...""" for readable multiline text blocks.

fmt **Formatting**

Format output so text appears clean and easy to scan.

```
Multiline examples
>>> message = "Hello\nPython"
>>> print(message)
Hello
Python
>>> poem = """Line one
... Line two"""
```

Concept	Best use	Reminder
Concatenation	Join two or more strings.	Add spaces yourself.
Indexing	Read one character.	Counting starts at 0.
Slicing	Extract a substring.	End index is excluded.
Immutability	Understand safe updates.	Create a new string.

Day 08/55 takeaway

Build text with concatenation, read with indexes, extract slices, and format output clearly. Strings are immutable, so edits make new strings.